Ide x AI '26

project

DESIGN MILESTONES

Coffee machine for offices and shared spaces

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VISUAL DESIGN MILESTONES AND MY REFLECTIONS THROUGHOUT DESIGNIN PROCESS

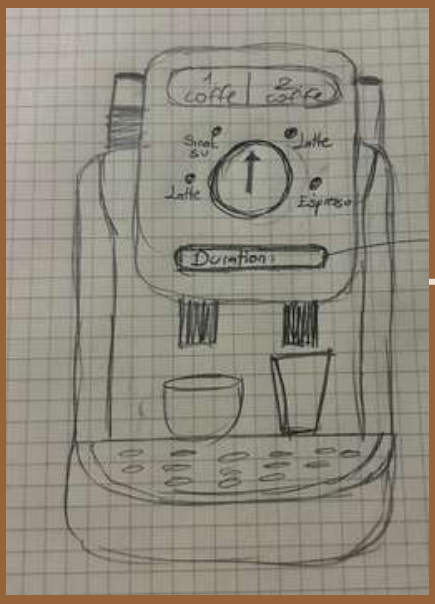
1

Promt: "Render this sketch."

2

-After seeing the results of the render and doing some online research, I realized there were many similar coffee machines like this. So, I drew a new sketch and reduced the coffee options because the internal mechanics were becoming too complex and significantly increasing the costs.-

3



Promt: "Render this sketch."

4

"After this render, I felt that the pipe coming out of the milk tank looked unprofessional. So, I decided to integrate the milk tank into the machine and design it with a 'plug-and-play' drawer mechanism." I draw antoher sketch and I rendered it again.

5

In this render,there were no coffee brewer so I ask gemini to add coffee brewer on top instead of water tank, I decided to put water tank milk tank inside of the coffee machine. I also add a button in the touch screen for user selection.

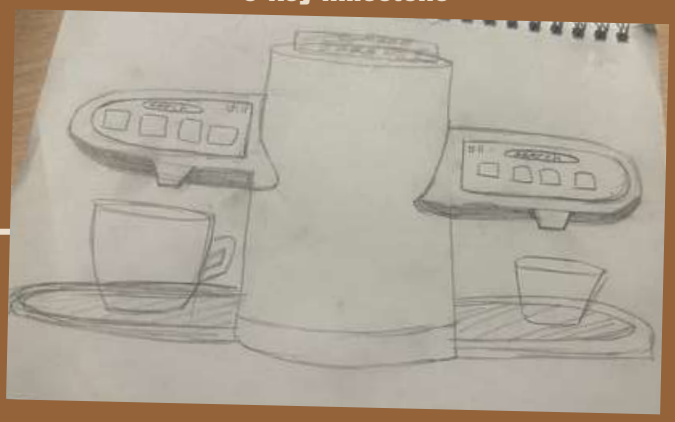
6

In this render, reshaped the milk tank and made machine more geometric structured. I also add an feature of milk

7

I added a feature that allows users to adjust the milk and coffee intensity on a 0–100 scale after selecting their drink. I asked AI to render this idea, and although the design slightly changed during the rendering process, the new layout still felt logical and functional. Because of this, I decided to keep the updated design.

8-key milestone



After receiving critique, I started thinking about how to make the design language more focused on “two-user interaction.” Instead of keeping the classic rectangular coffee machine form, I decided to redesign the shape and created this sketch. I moved the brewing arms further apart so that two users could comfortably take their coffee at the same time. I also increased the number of screens to two, allowing both users to easily select their drinks simultaneously. And I rendered it.

9

Promt: "Spread the arms to the sides and position the two touchscreens with a slight tilt towards the user."

Why this promt has given ?

The touchscreens are tilted to provide better ease of use for the user. When positioned completely upright, the user may find it difficult to see or touch the screen. However, when angled slightly toward them, it becomes much easier to view and interact with

10

Promt: "Make 2 coffees, not 4."

Why this promt has given ?

"Making four coffees in a single machine is nearly impossible at these dimensions, and even if it were achieved, it wouldn't be functional. From the very beginning, this machine was designed to brew two coffees.

11

Promt: "Change the metal arms to red plastic and add a coffee grinder on top."

Why this promt has given ?

"Making four coffees in a single machine is nearly impossible at these dimensions, and even if it were achieved, it wouldn't be functional. From the very beginning, this machine was designed to brew two coffees.

12

Promt: “Remove the milk tank from the middle."

Why this promt has given ?

At this stage, the product reached the form I wanted. However, having the milk visible in a glass tube in the center did not seem logical, since the machine already includes a removable milk tank inside. Therefore, I prompted AI to remove the central milk tube to make the design more realistic and consistent with the internal system.

11- final product

DESIGN MILESTONES FOR TOUCH SCREEN AND THE FEATURES

1. Interface Shape Development

At the beginning, the machine had a small circular touchscreen interface. However, this format limited the readability of drink options and interaction space. Therefore, the interface was redesigned into a more square-shaped touchscreen, which allowed clearer navigation and better visibility for users selecting their coffee.

2. Simplifying Coffee Options

Initially, the machine offered around ten different coffee options. After further consideration, this number was reduced to four core drinks. This decision made the interface simpler for users and also improved the mechanical feasibility of the machine, since supporting many drink variations would complicate the internal brewing system.

3. Coffee & Milk Intensity Adjustment Feature

To increase personalization, a new feature was added that allows users to adjust the coffee strength and milk intensity on a 0–100 scale. This feature gives users more control over their drink while keeping the menu itself simple and easy to navigate.

4. Personalized Coffee Profiles

Finally, a profile feature was introduced where users can save their preferred coffee settings by entering their name. When they arrive and select their profile, the machine automatically prepares their personalized drink, making the system more convenient for shared environments such as offices.

What I accepted, rejected and transformed from AI

Throughout the design process, I both accepted and rejected several ideas suggested by AI while transforming others into more realistic solutions. I accepted suggestions such as the dual-path brewing system, Smart Manifold water distribution, and linear packaging layout, because they made the simultaneous coffee preparation concept technically feasible within a compact structure. However, I rejected ideas like vertical expansion, overly complex mechanisms, and integrated sugar dispensing, since they conflicted with the goal of keeping the machine compact and maintainable for shared environments. Instead, I refined and adapted some AI ideas—such as shifting the design to a horizontal layout and limiting drink options—to create a more realistic and implementable system.

some rejected renders:

Selected Impactful Prompts

These prompts significantly influenced the design direction:

“bence en mantıklı çözüm kahve kapasitesini ikiye çıkarmak.”
→ Led to the core design decision: dual brewing architecture

“bence dikeyde değil yatayda genişletelim.”
→ Changed product geometry toward horizontal system layout

“iki kişi farklı kahveleri yapabilsin.”
→ Introduced independent brewing paths

“4 mü 6 kahve mi daha uygun olur.”
→ Led to menu optimization and system efficiency

“aynı anda iki kahve gerçekten üretilebilir mi?”
→ Clarified realistic engineering limitations.

“su ve süt deposunu nasıl yerleştirebilirim.”
→ Triggered central cylinder storage architecture

Selected Keywords

CoreSync
Shared living systems
Dual-path coffee system
Shared resource management
Compact appliance design
Human–AI co-design
User conflict reduction

AI Role Tags

Clarifier
AI helped clarify user behavior patterns, coffee brewing processes and engineering feasibility.
Example:
Explaining thermoblock logic and pressure systems.

Re-framer
AI helped reframe the design problem from: “single coffee machine design” to “shared resource system in collective living environments.”

Expander
AI expanded design possibilities by generating concept alternatives, system features and engineering solutions.
Example:
Conflict-aware brewing logic, modular upgrades, acoustic sensing.

Critique
AI also questioned feasibility thermal crowding risks, simultaneous grinding limitations, maintenance complexity. This helped refine the concept toward more realistic engineering logic.

My Reflections

Throughout the process, working with AI helped me shift my thinking from designing only the form of a coffee machine to understanding it as a system that combines user interaction, internal mechanics, and shared use scenarios. The conversations clarified important aspects such as simultaneous brewing logic, heating systems, and how internal components like water and milk tanks could be organized within the central structure. It also helped me reflect on how interface decisions—such as simplifying drink options and adding adjustable intensity—can make the product both technically feasible and easier for users to interact with.

My Design Decisions

During the design process, I made several key decisions by evaluating and refining AI suggestions. I accepted ideas that improved the technical logic of the product, such as the dual brewing concept, simplified drink menu, and adjustable coffee and milk intensity. At the same time, I rejected elements that did not align with the goal of creating a compact and realistic shared-use machine, such as overly complex mechanical solutions or unnecessary visual features. I also transformed some ideas—for example redesigning the machine form to better support two simultaneous users, adding dual touchscreens, and introducing personalized coffee profiles to enhance usability in shared environments.